

3D printing exercise manual

Digital Design and Fabrication (inf175)

Media Informatics and Multimedia Systems
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Today's Task

In today's exercise session, you will:

1. Design and 3D model a **practical object**.
2. Import it into slicing software.
3. Decide on the **optimal orientation** for printing.
4. Enable and adjust **supports** (and/or a brim) if required.

Getting Help During the Session

Tutor(s) will be available throughout the session to help with any questions about parametric 3D modelling in Onshape and slicing in QIDI Studio. If you feel confident with the task and don't need any assistance, you're free to leave the exercise session at any point.

Theme: Smartphone Accessories

The default theme for today is **smartphone accessories**. This can be as simple as a stand for your phone, or something more creative, such as:

- A passive speaker horn
- A diffuser that turns your phone's flash into a camping lantern

For inspiration, take a look at the Printables phone stand contest:

[Phone and Tablet Stands | Contest theme | Printables.com](#)

You are **not limited to this theme**. You're free to design anything you can find a practical use for. You can find a few more sources of inspiration on the next slide.

A few more sources of inspiration:

[Practical models | Contest theme | Printables.com](#)

[AirTag Accessories | Contest theme | Printables.com](#)

[Keychain Tools | Contest theme | Printables.com](#)

[Shoe Hacks | Contest theme | Printables.com](#)

[Hair Accessories | Contest theme | Printables.com](#)

[Paper Towel Holders | Contest theme | Printables.com](#)

[Office Accessories | Contest theme | Printables.com](#)

[3D Design Contests | Printables.com](#)

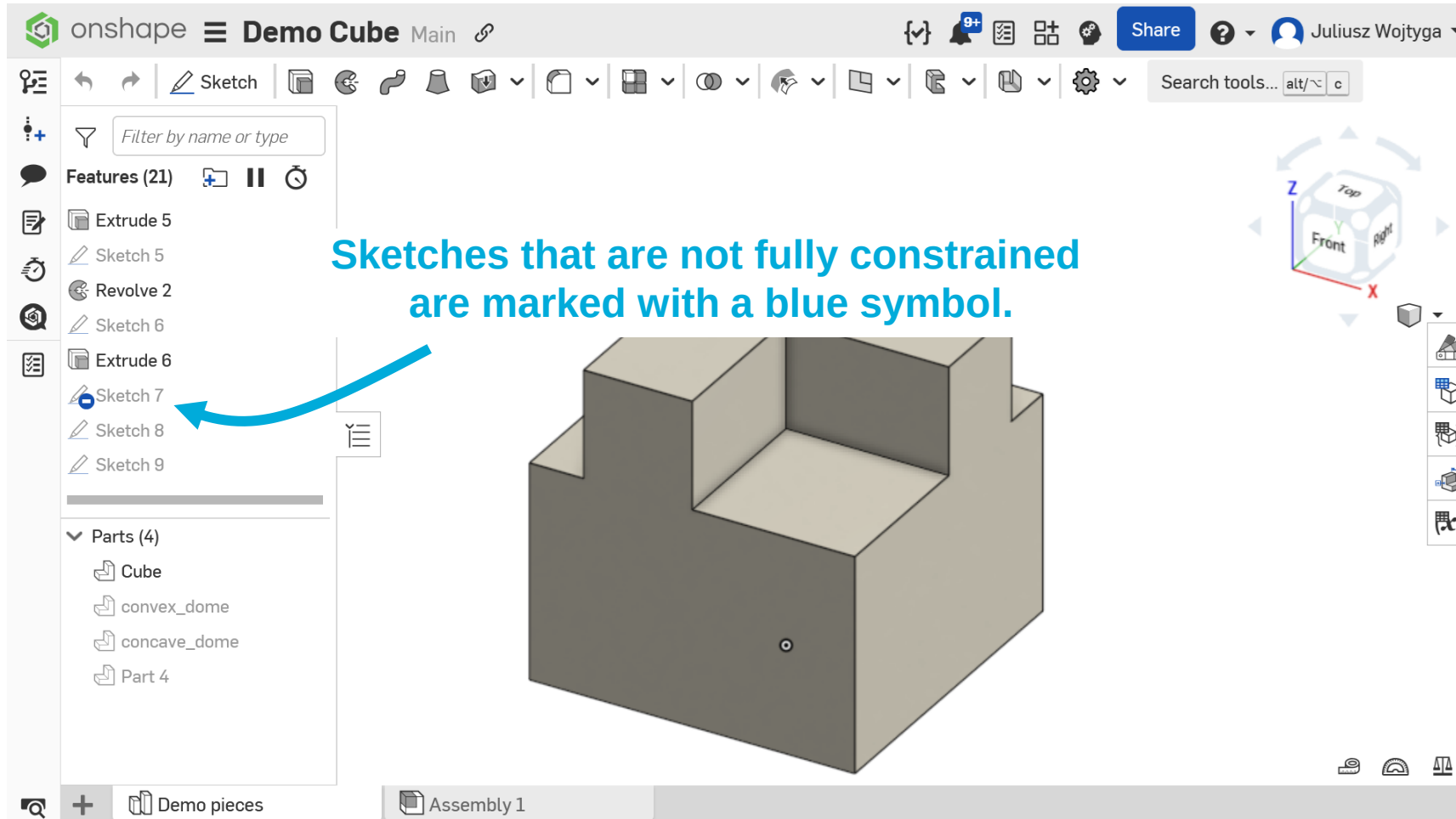
General Design Rules

- Your design can be a **single object** or **multiple parts** intended for later assembly.
- It must fit within the available printer's bed size — the **QIDI Q2** has a build volume of **270 × 270 mm** (bed) and **256 mm** of height.
- After slicing — including any supports — the model must **not exceed 120 g** of filament usage.
- By default, parts will be printed in **black PLA**. Availability of other colours and materials will be announced during class.

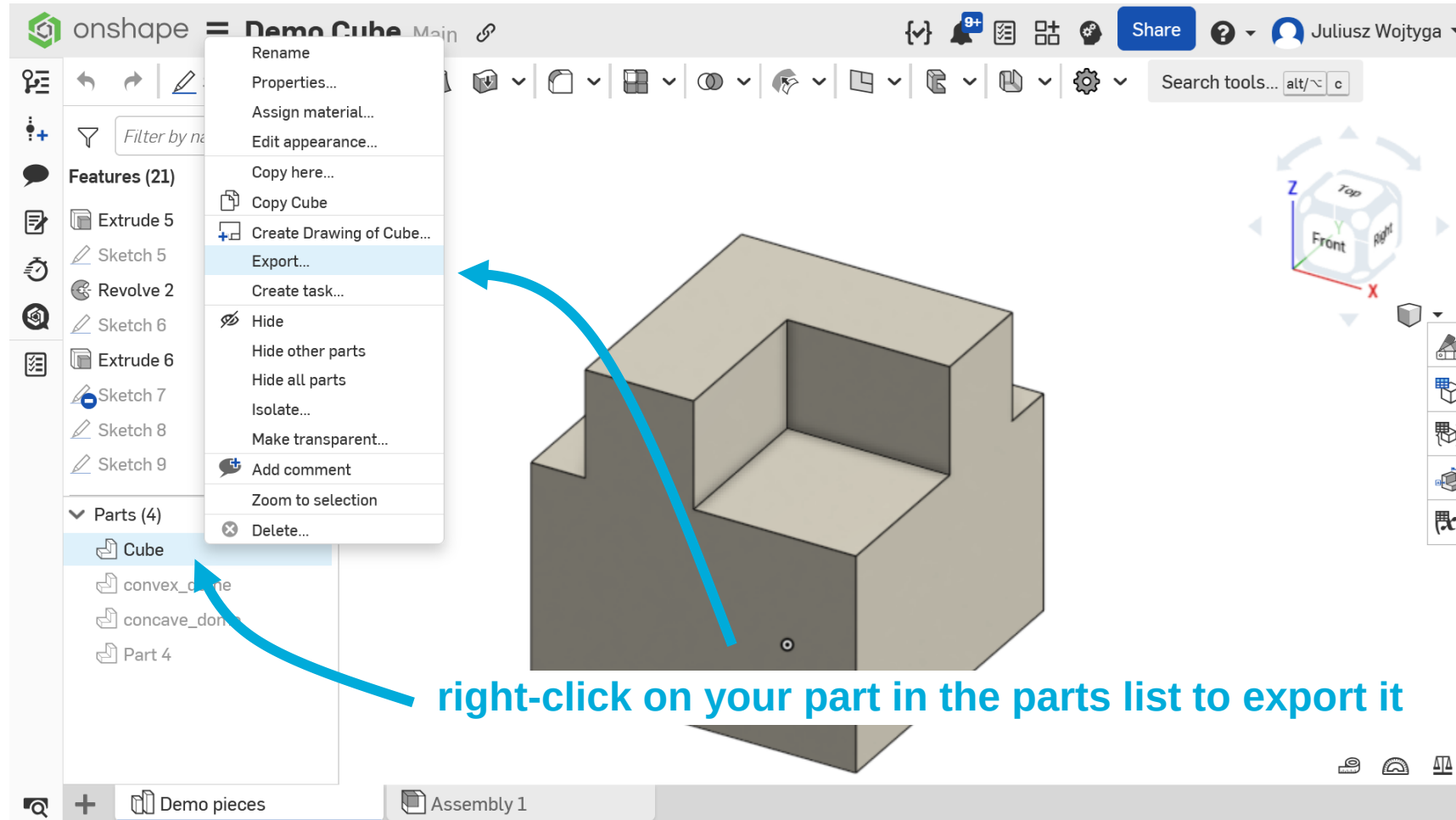
Part 1 – Onshape (CAD Modeling)

- Completing the **Onshape tutorials** is a prerequisite for this exercise. We will not provide assistance with Onshape unless proof of tutorial completion has been uploaded as instructed.
- Your Onshape document must be located in the **Digital Design and Fabrication (inf175)** class.
 - If you created your Onshape account with your university email (as instructed), you should already be added to the class.
 - If you are not, contact the tutor at the **beginning** of the exercise session.
- Make sure your Onshape units are set to **millimetres**.
- **All sketches** that make up your part **must be fully constrained**.

Part 1 – Onshape (CAD Modeling)



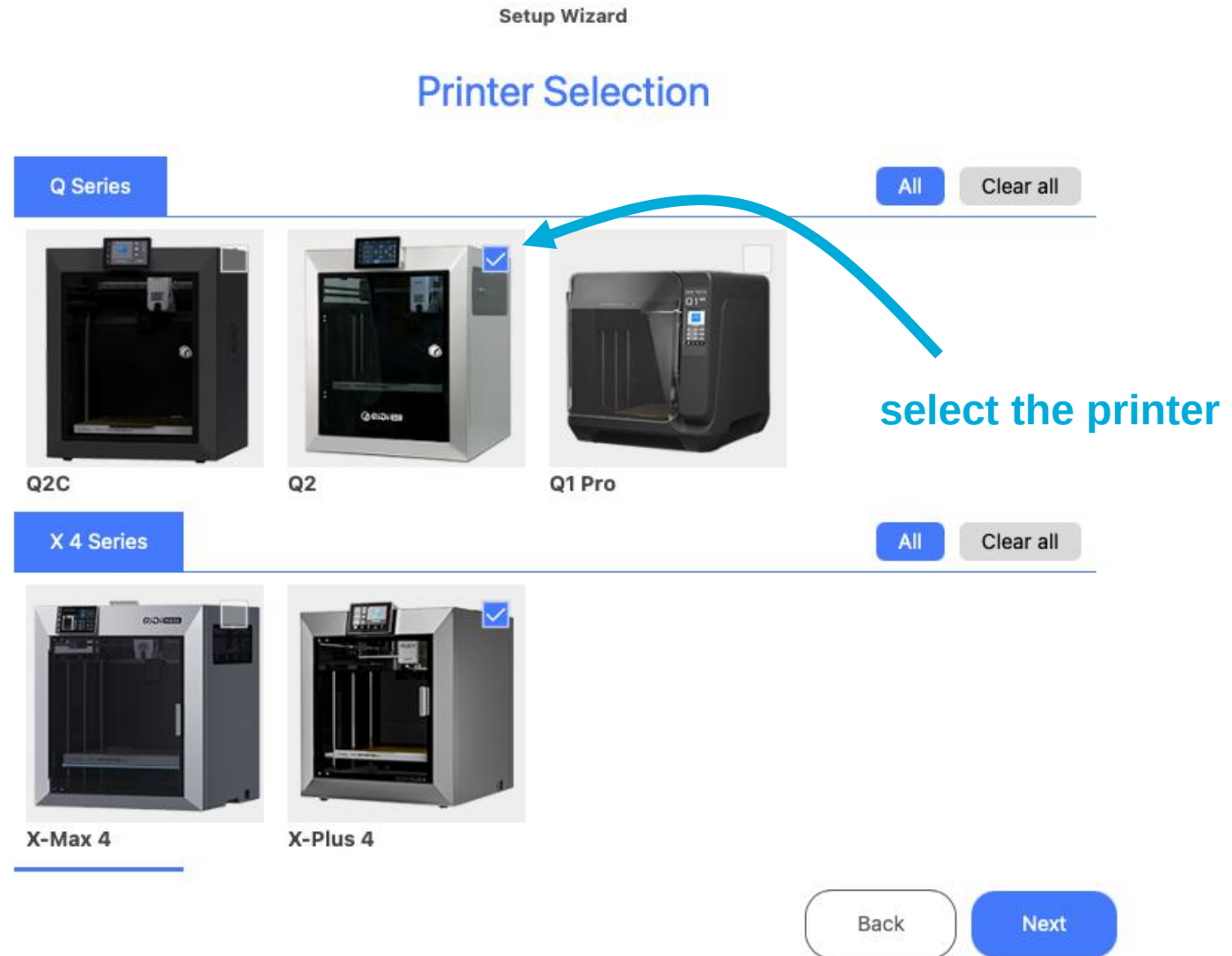
Part 1 – Onshape (CAD Modeling)



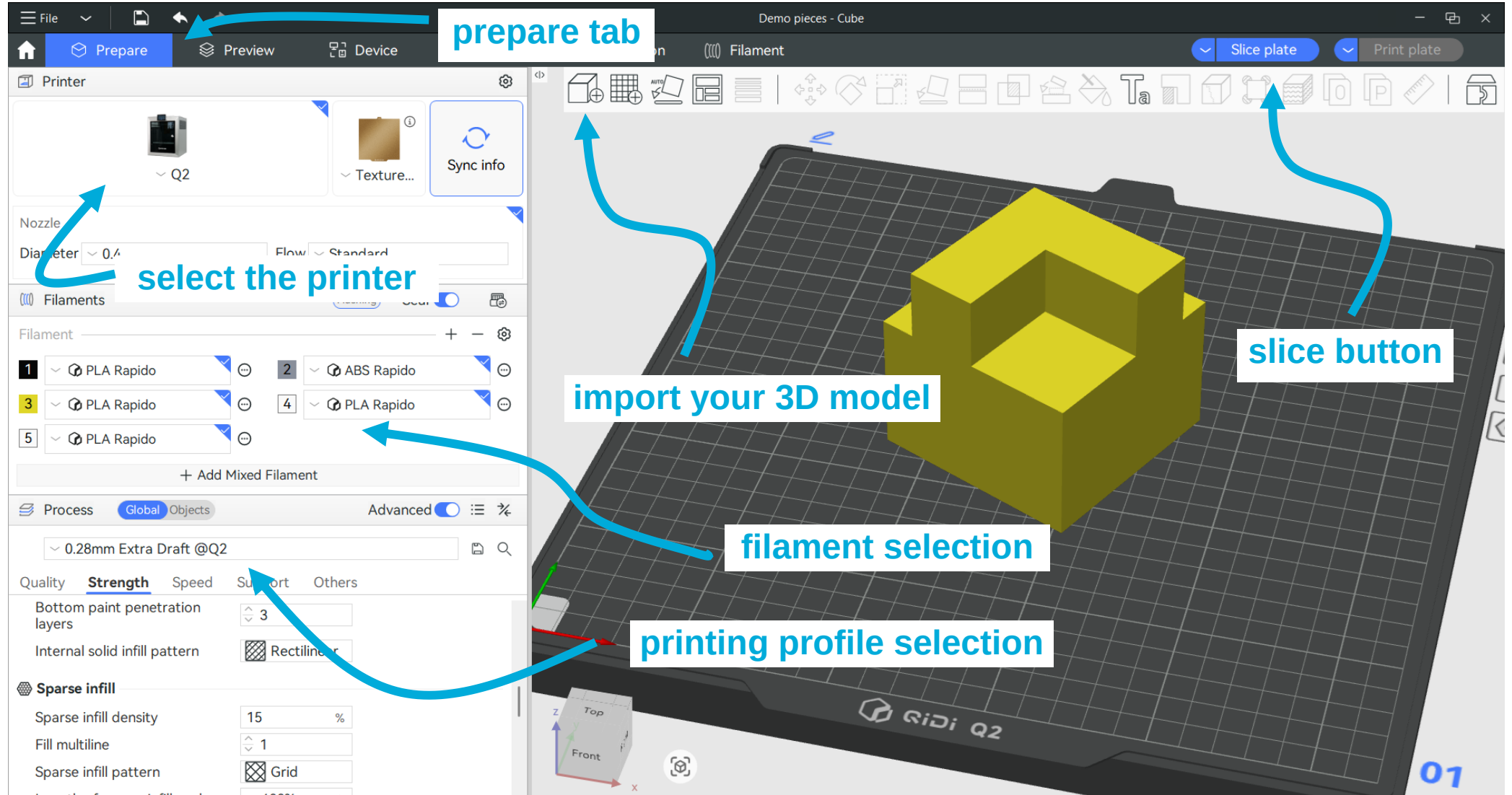
Part 2 – Slicer (QIDI Studio)

- Download **QIDI Studio** [here](#). During setup, select **Q2** as your printer.
- Import your model into the slicer:
 - Select the appropriate printing profile.
 - Select PLA Rapido as your filament.
 - Orient the model in space.
 - Enable supports and/or a brim if needed.
 - Slicing must complete **without any errors**.
- You're free to adjust other print parameters, such as infill, wall count, top/bottom shell thickness, etc.
- Once successfully sliced, **save the project as a 3MF file**.

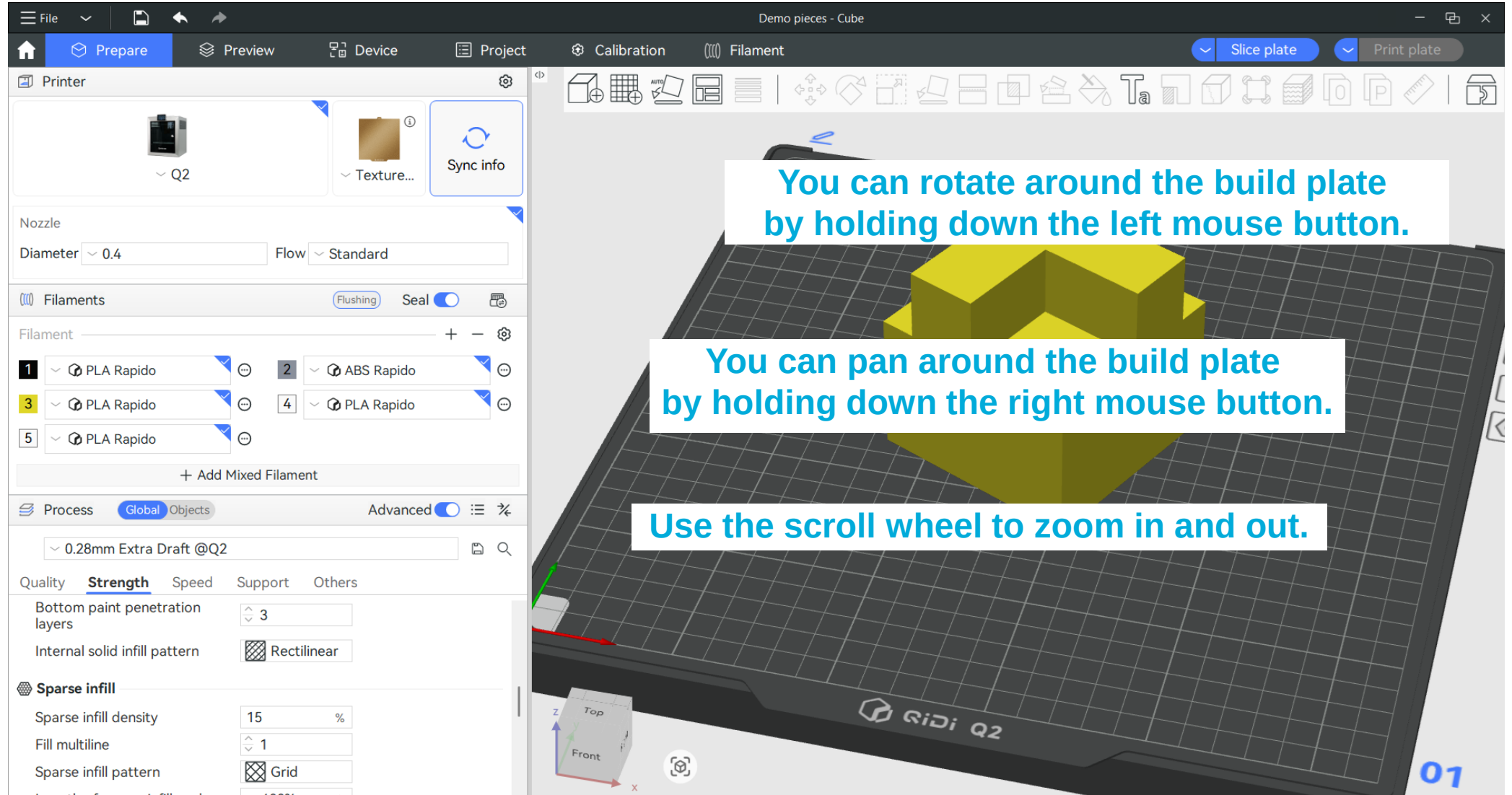
How to use a slicer: the basics



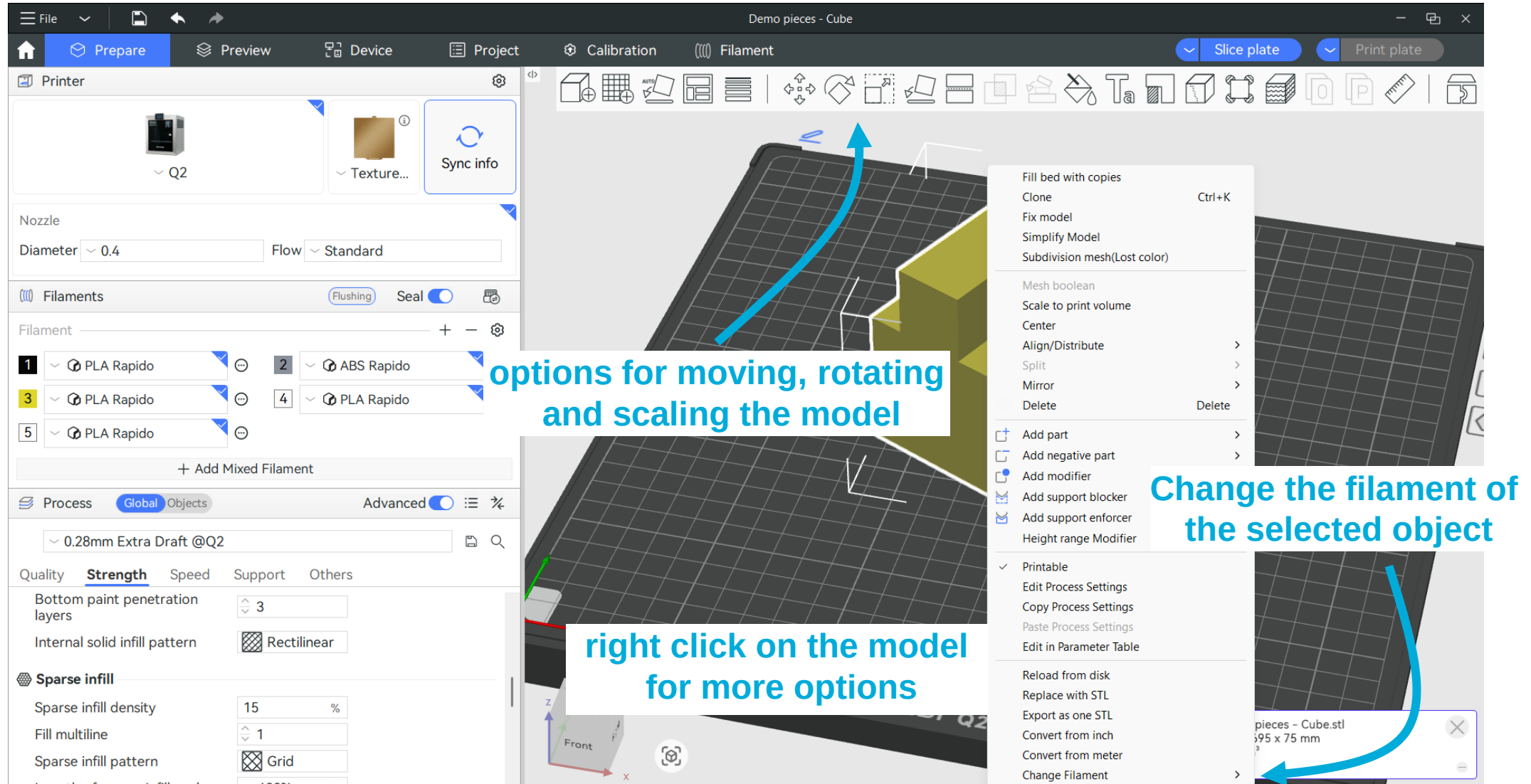
How to use a slicer: the basics



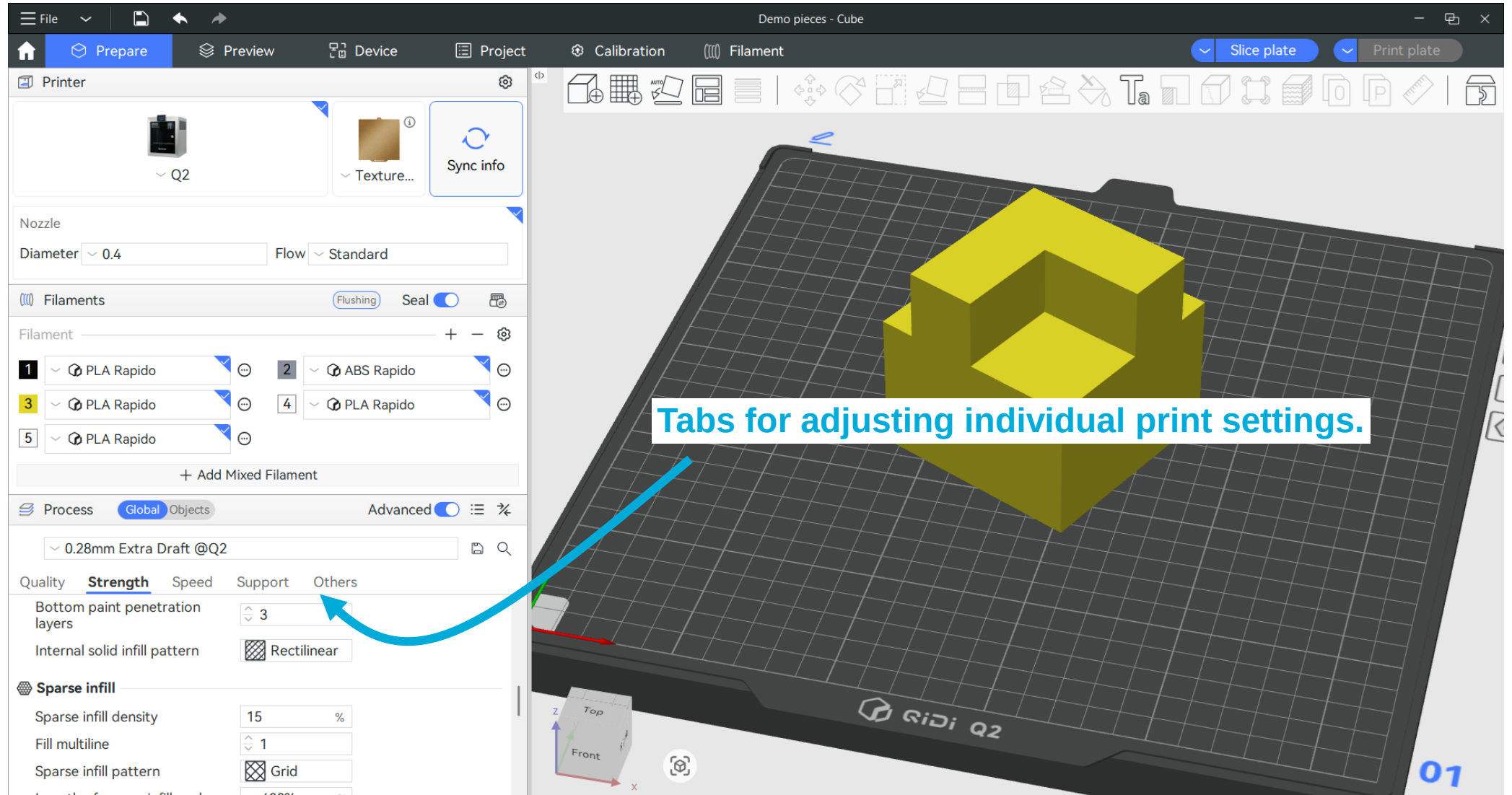
How to use a slicer: the basics



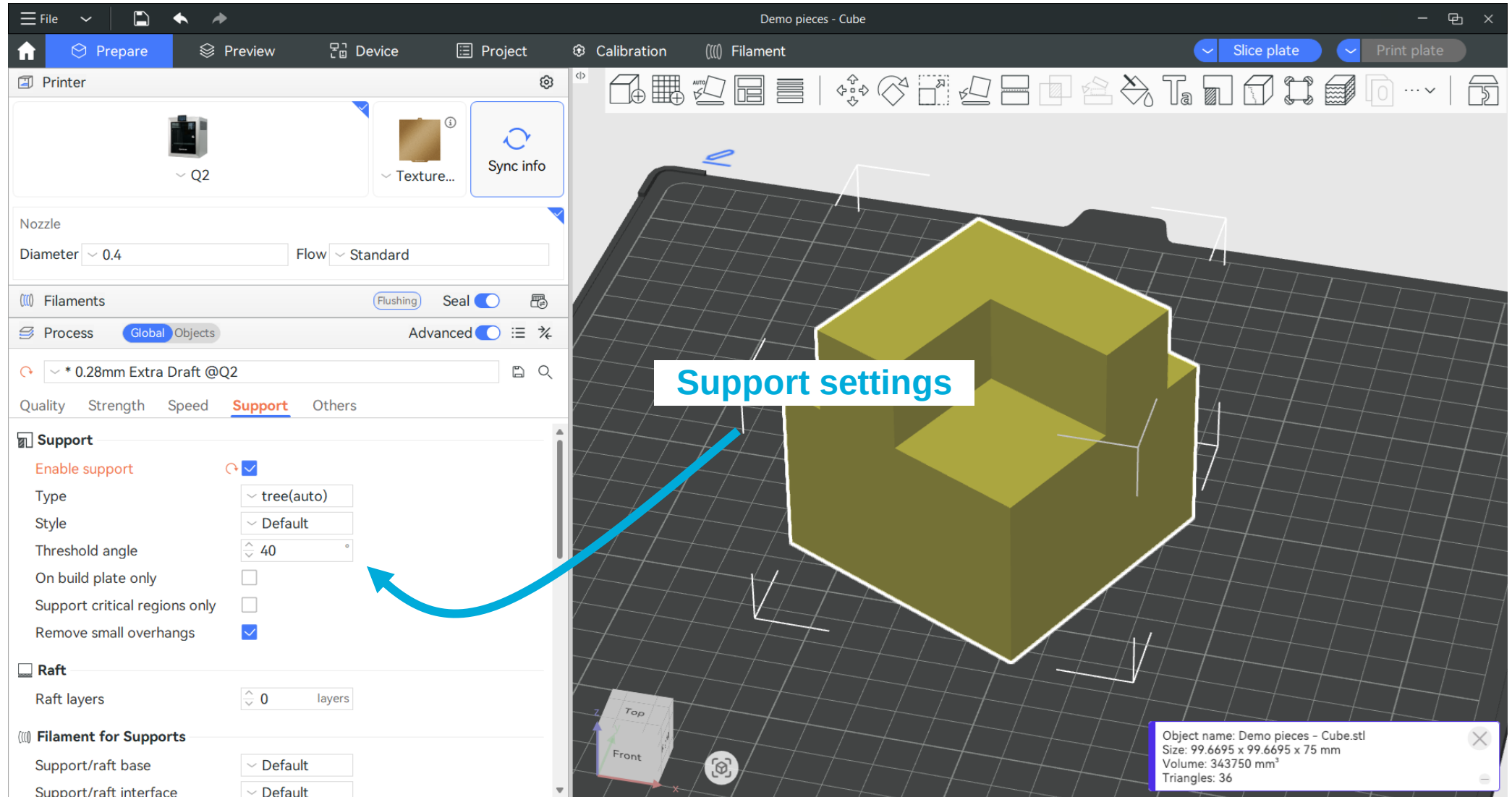
How to use a slicer: the basics



How to use a slicer: the basics



How to use a slicer: the basics



Bed adhesion — brim settings in QIDI slicer

Brim type: controls how the brim is generated — Auto lets QIDI Studio automatically analyse each object and generate an appropriately sized brim, based on the part's shape/orientation, brim type, width, and gap. The other options are:

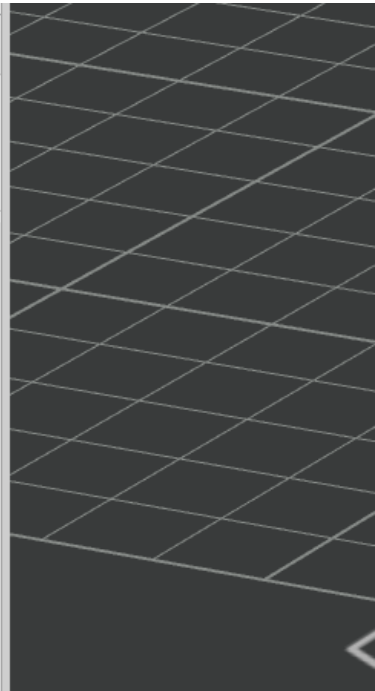
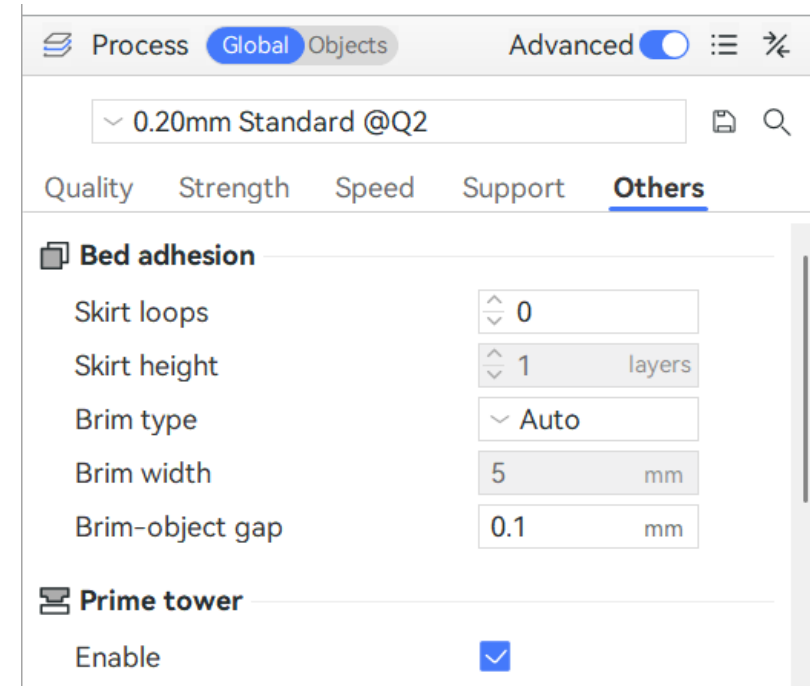
Outer brim only — generates brim only around the outside of the object.

Inner brim only — if the object is hollow, generates brim only around the inside of the object.

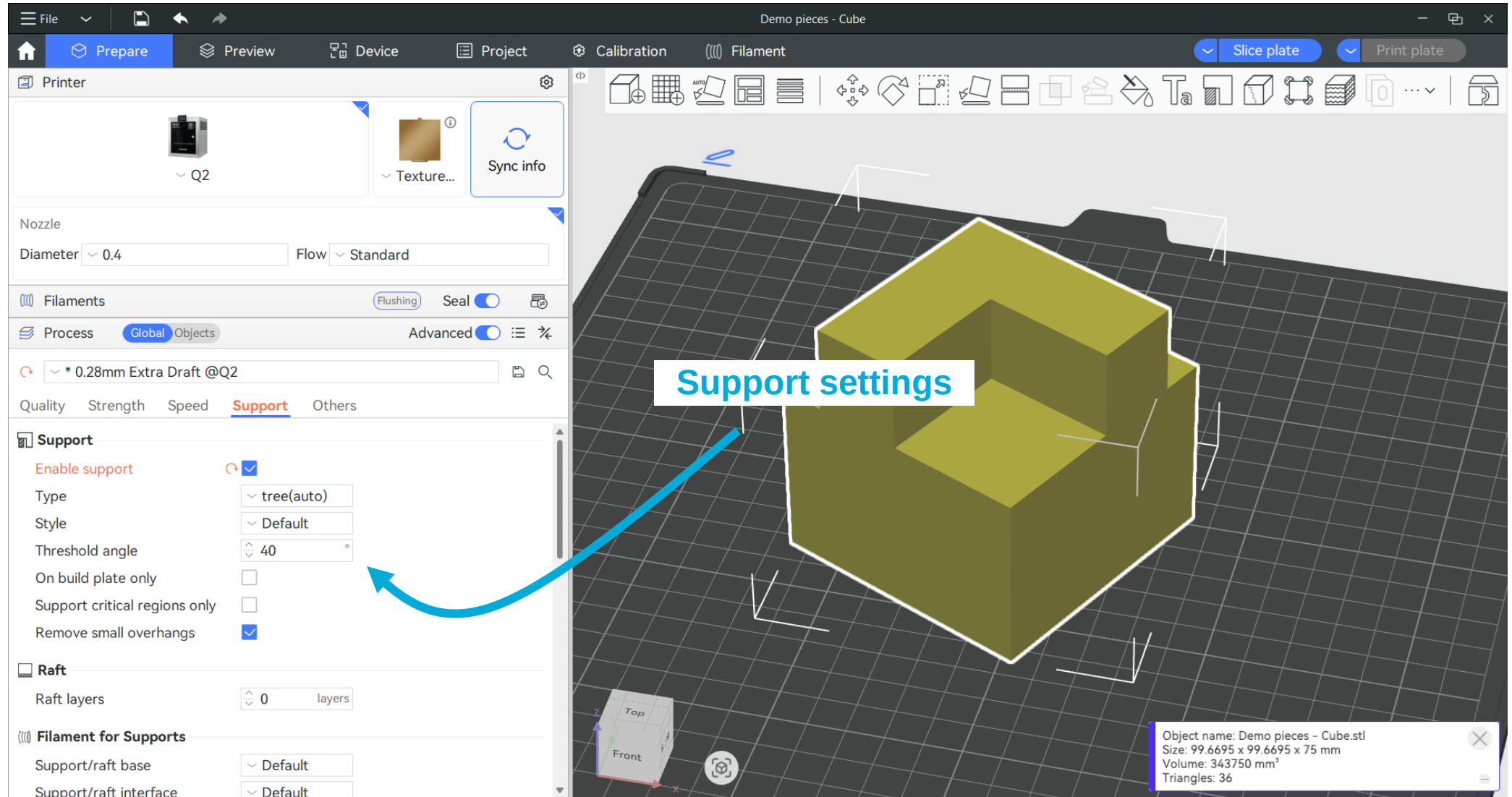
Outer and inner brim — generates brim around both the inside and outside of the object.

No-brim — no brim is generated at all.

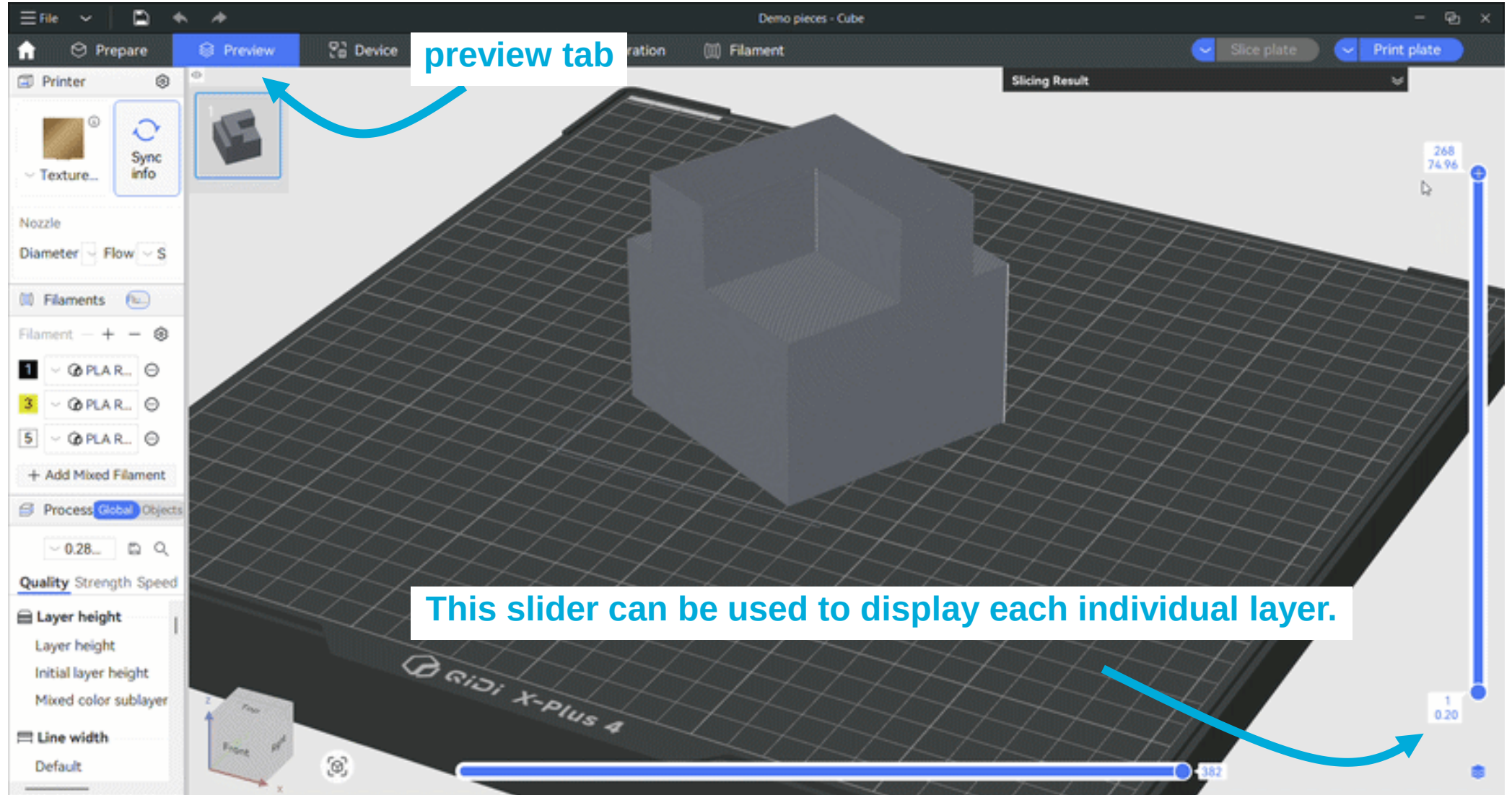
Brim width: only manually editable once Brim type is set to something other than Auto. Wider brim = more bed contact and holding power, at the cost of more material and cleanup.



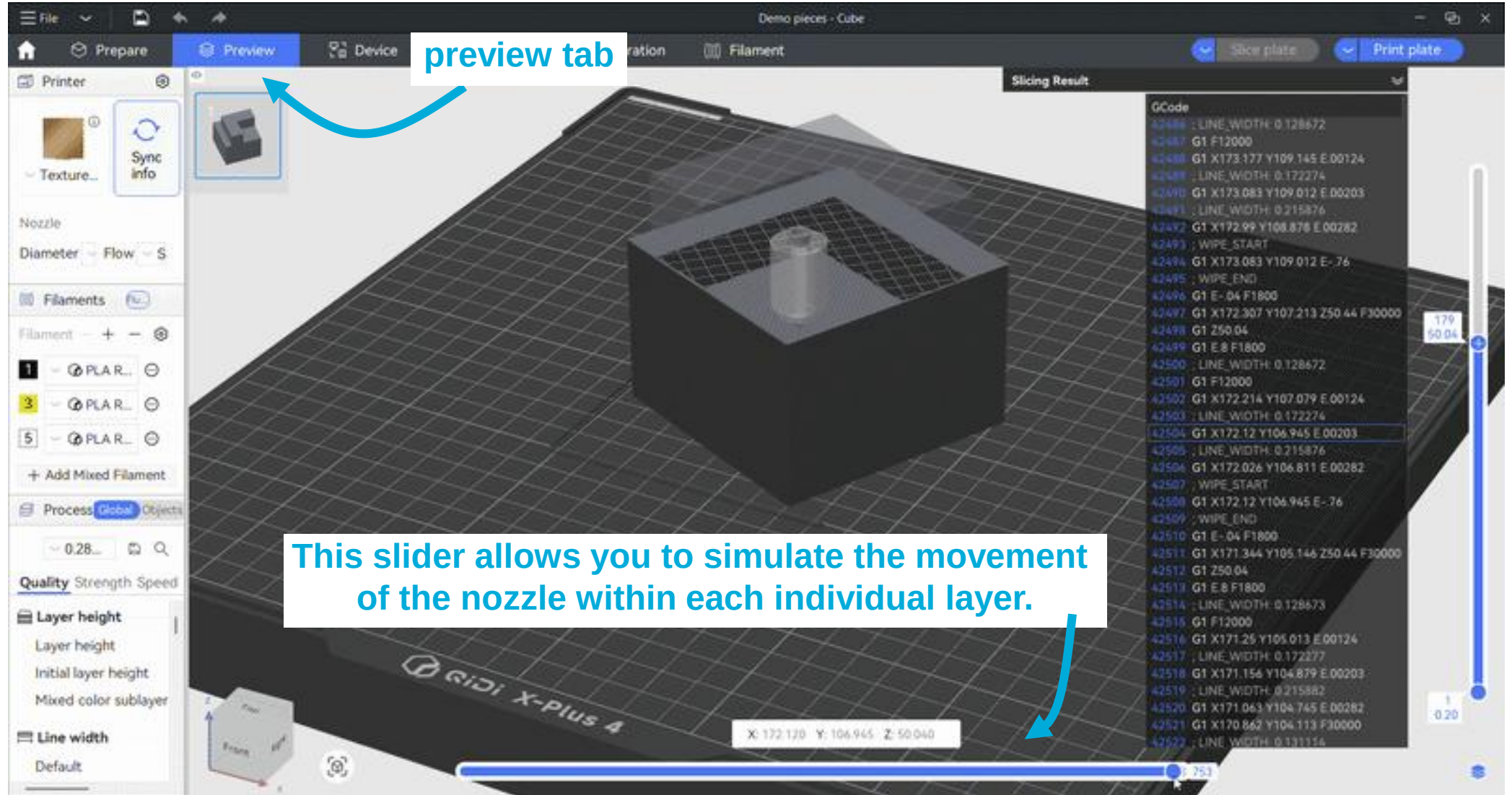
How to use a slicer: the basics



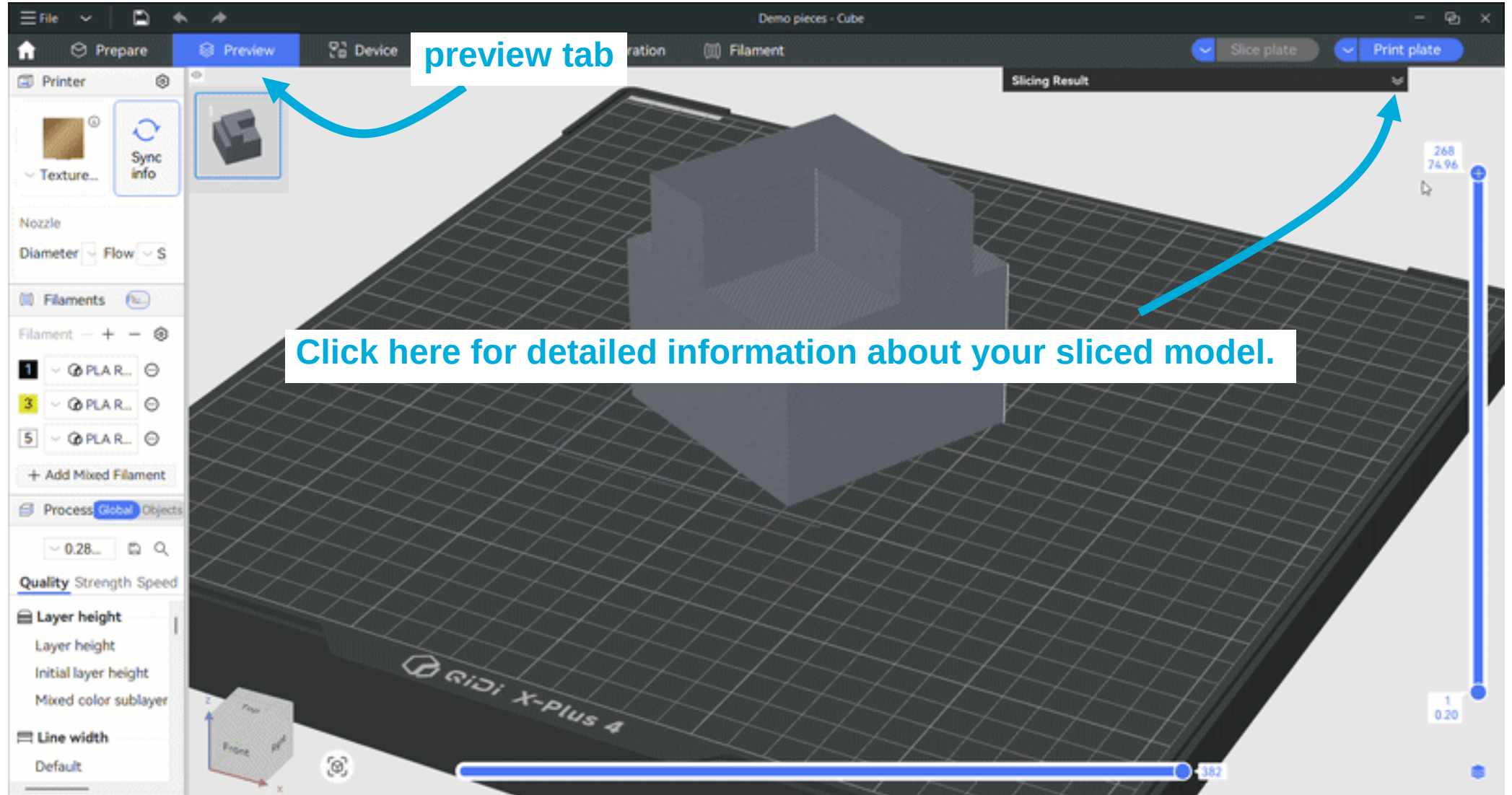
How to use a slicer: the basics



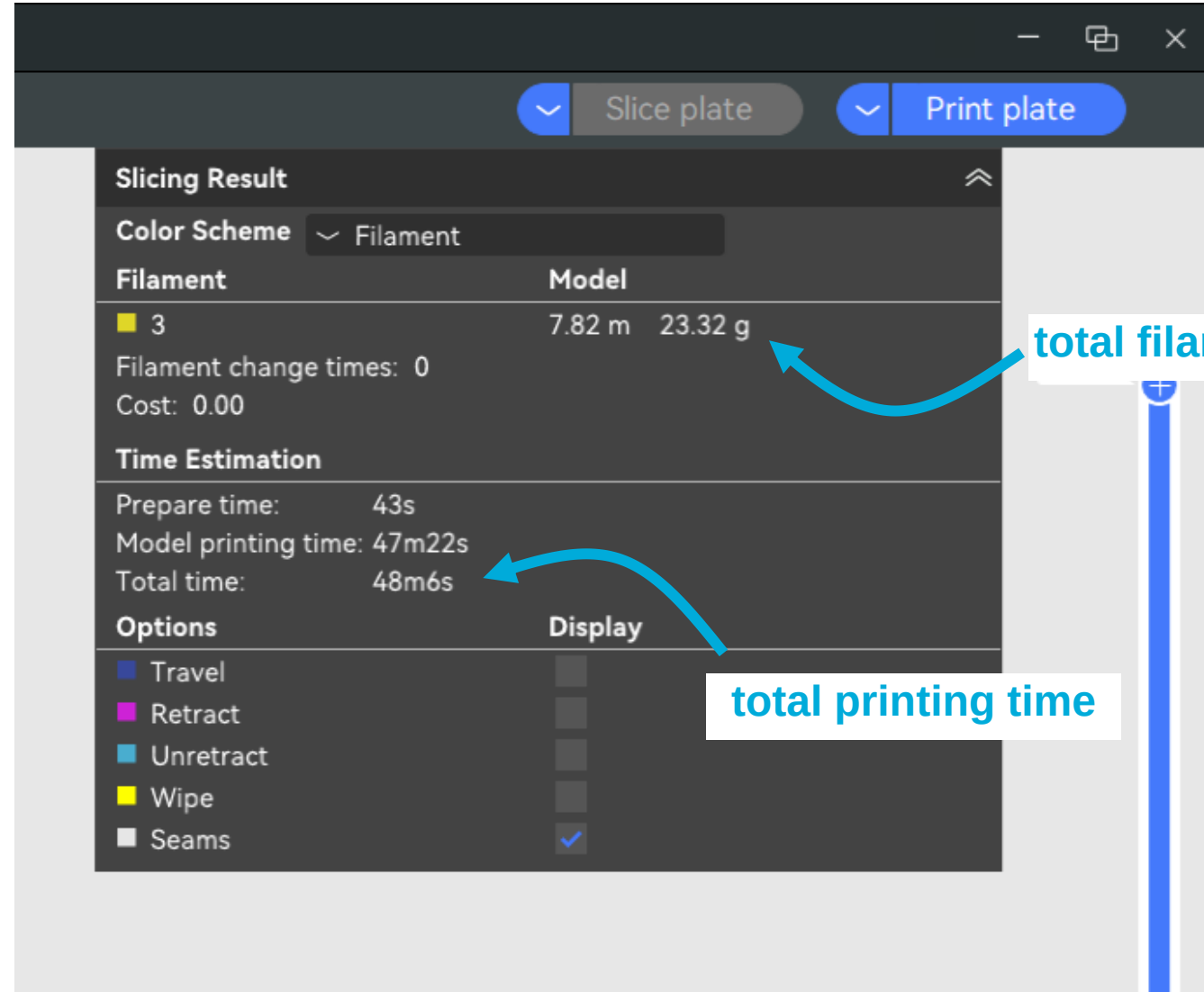
How to use a slicer: the basics



How to use a slicer: the basics



How to use a slicer: the basics



total filament used in grams

total printing time

Submission Requirements — Due 30.06.2026

You are required to upload the following two files:

1. The exported **STEP file(s)** from Onshape
2. The saved slicer **project file (.3mf)**

Where to upload: Stud.IP, in the folder "**3D printing exercise - 3D model .step and .3mf slicer project files.**"

Naming requirements:

- Both files must have a **descriptive name** and include your **full name**.
 - Example: *phone_stand_Juliusz_Wojtyga.step*
 - Files named like *Part Studio 1 - Part 1.step* will be **rejected**.

Printed Models & Portfolio Deadline

Your designs will be printed by the teaching team after submission. The printed models will be **distributed during next week's class (02.07.2026)**.

Once your printed model has been distributed to you, the deadline for the **portfolio entry** for this exercise is:

Tuesday, 07.07.2026, 16:00

Portfolio

As usual, include a description of your process covering:

- What did you decide to model, and why? How do you intend to use your printed accessory? Does it solve any problem?
- Is this your first time using parametric CAD?
- How did you find the tutorials? Were there any challenges?
- How difficult was it to turn your idea into a 3D model?
- Which slicer settings did you apply, and in what orientation did you slice your model?
- Pictures of the final, printed product